

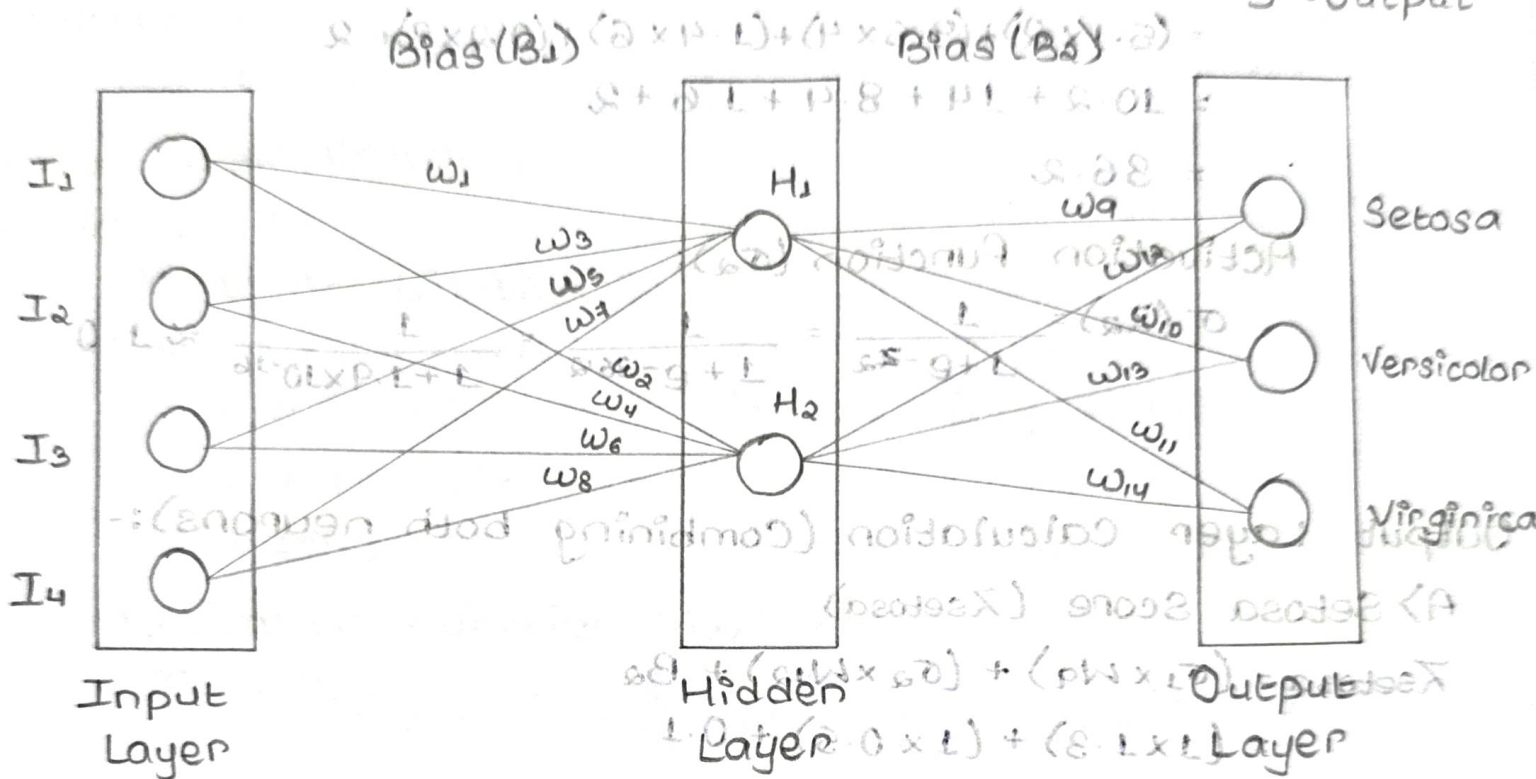
Assignment - 2

Neural Network Forward Propagation Using Mathematical Calculations

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Mathematical Walkthrough of a Single-Hidden-Layer Neural Network using the Iris Dataset:

4 → Input
3 → Output



Part 1: Calculation for Row 1

Input Vector: $X = [5.1, 3.5, 1.4, 0.2]$

Fixed Model Parameter:-

Hidden Weights (H_1): $w_1 = 1, w_3 = 3, w_5 = 5, w_7 = 7$

Hidden Weights (H_2): $w_2 = 2, w_4 = 4, w_6 = 6, w_8 = 8$

Output Weights (from H_1): $w_9 = 1.3, w_{10} = 0.5, w_{11} = 1.2$

Output Weights (from H_2): $w_{12} = 0.5, w_{13} = 0.2, w_{14} = 0.4$

Biases: $B_1 = 2, B_2 = 0.1$

Hidden Layer Calculation:-

A) Neuron 1:

Linear combination (Z_1):

$$\begin{aligned} Z_1 &= X_1 w_1 + X_2 w_3 + X_3 w_5 + X_4 w_7 + B_1 \\ &= (5.1 \times 1) + (3.5 \times 3) + (1.4 \times 5) + (0.2 \times 7) + 2 \\ &= 5.1 + 10.5 + 7 + 1.4 + 2 \Rightarrow 26 \end{aligned}$$

Activation Function (σ_1):

$$\sigma(z_1) = \frac{1}{1+e^{-z_1}} = \frac{1}{1+e^{-26}} = \frac{1}{1+5.11 \times 10^{-12}} \approx 1.0$$

B) Neuron 2:

Linear Combination (z_2):

$$\begin{aligned} z_2 &= x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + b_1 \\ &= (5.1 \times 2) + (3.5 \times 4) + (1.4 \times 6) + (0.2 \times 8) + 2 \\ &= 10.2 + 14 + 8.4 + 1.6 + 2 \\ &= 36.2 \end{aligned}$$

Activation Function (σ_2):

$$\sigma(z_2) = \frac{1}{1+e^{-z_2}} = \frac{1}{1+e^{-36.2}} = \frac{1}{1+1.9 \times 10^{-16}} \approx 1.0$$

Output Layer Calculation (Combining both neurons):-

A) Setosa Score (z_{setosa})

$$\begin{aligned} z_{\text{setosa}} &= (\sigma_1 \times w_9) + (\sigma_2 \times w_{12}) + b_2 \\ &= (1 \times 1.3) + (1 \times 0.5) + 0.1 \\ &= 1.9 \end{aligned}$$

B) Versicolor Score ($z_{\text{versicolor}}$)

$$\begin{aligned} z_{\text{versicolor}} &= (\sigma_1 \times w_{10}) + (\sigma_2 \times w_{13}) + b_2 \\ &= (1 \times 0.5) + (1 \times 0.2) + 0.1 \end{aligned}$$

$$F = 0.8$$

C) Virginica Score ($z_{\text{virginica}}$)

$$\begin{aligned} z_{\text{virginica}} &= (\sigma_1 \times w_{11}) + (\sigma_2 \times w_{14}) + b_2 \\ &= (1 \times 1.2) + (1 \times 0.4) + 0.1 \\ &= 1.7 \end{aligned}$$

Row 1 Final Output Vector: [1.9, 0.8, 1.7]

Prediction: Setosa (Highest score of 1.9)

Part 2: Calculation for Row 2

Input Vector: $X = [4.9, 3.0, 1.4, 0.2]$

Fixed Model Parameters:-

Hidden Weights (H_1): $W_1 = 1, W_3 = 3, W_5 = 5, W_7 = 7$

Hidden Weights (H_2): $W_2 = 2, W_4 = 4, W_6 = 6, W_8 = 8$

Output Weights (from H_1): $W_{10} = 0.5, W_{11} = 1.2$

Output Weights (from H_2): $W_{12} = 0.5, W_{13} = 0.2, W_{14} = 0.4$

Biases: $B_1 = 2, B_2 = 0.2 \times 0.1 + (2.0 \times 0.1) =$

Hidden Layer Calculation:-

A) Neuron 1:

Linear Combination (z_1):

$$\begin{aligned} z_1 &= x_1 W_1 + x_2 W_3 + x_3 W_5 + x_4 W_7 + B_1 \\ &= (4.9 \times 1) + (3.0 \times 3) + (1.4 \times 5) + (0.2 \times 7) + 2 \\ &= 4.9 + 9.0 + 7.0 + 1.4 + 2 \\ &= 24.3 \end{aligned}$$

Activation Function (σ):

$$\sigma(z_1) = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-24.3}} = \frac{1}{1 + 2.80 \times 10^{-11}} \approx 1.0$$

B) Neuron 2:

Linear Combination (z_2):

$$\begin{aligned} z_2 &= x_1 W_2 + x_2 W_4 + x_3 W_6 + x_4 W_8 + B_2 \\ &= (4.9 \times 2) + (3.0 \times 4) + (1.4 \times 6) + (0.2 \times 8) + 2 \\ &= 9.8 + 12.0 + 8.4 + 1.6 + 2 \\ &= 33.8 \end{aligned}$$

Activation Function (σ):

$$\sigma(z_2) = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-33.8}} = \frac{1}{1 + 2.09 \times 10^{-15}} \approx 1.0$$

Output Layer calculation (Combining both neurons):

A) Setosa Score (Z_{setosa})

$$\begin{aligned} Z_{\text{setosa}} &= (\sigma_1 \times W_{1a}) + (\sigma_2 \times W_{1b}) + B_a \\ &= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1 \\ &= 1.9 \end{aligned}$$

B) Versicolor Score ($Z_{\text{versicolor}}$)

$$\begin{aligned} Z_{\text{versicolor}} &= (\sigma_1 \times W_{1c}) + (\sigma_2 \times W_{1d}) + B_b \\ &= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1 \\ &= 0.8 \end{aligned}$$

C) Virginica Score ($Z_{\text{virginica}}$)

$$\begin{aligned} Z_{\text{virginica}} &= (\sigma_1 \times W_{1e}) + (\sigma_2 \times W_{1f}) + B_c \\ &= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1 \\ &= 1.7 \end{aligned}$$

Row 2 Final Output Vector: $[1.9, 0.8, 1.7]$

Prediction: Setosa (Highest score of 1.9)

$$0.1 \approx \frac{1}{0.1 \times 0.8 + 1} = \frac{1}{0.08 + 1} = \frac{1}{1.08} = (x)$$

Linear Combination (Z_2):

$$\begin{aligned} Z_2 &= x_1 W_2 + x_2 W_3 + x_3 W_4 + B_2 \\ &= 0.1(8) + (2 \times 1.1) + (1.4 \times 0.8) + (1.9 \times 0.1) \\ &= 0.8 + 2.2 + 1.12 + 0.19 = 4.31 \end{aligned}$$

Activation Function (σ):

$$0.1 \approx \frac{1}{0.1 \times 4.31 + 1} = \frac{1}{0.431 + 1} = \frac{1}{1.431} = (x)$$